

7.7 M-ary PAM (PAM-M) Transmission

A baseband binary PAM system is a pulse amplitude modulator that produces pulses that can have two amplitude levels, known as binary pulses. A baseband M-ary PAM system is similar to the binary PAM system, with the difference being that an M-ary PAM system can output M possible amplitude levels, with $M > 2$. Figure 7.11(a) shows a quaternary ($M = 4$) system outputting a binary data sequence 0010110111, while Figure 7.11 (b) shows the electrical representation for each bit pair. M-ary system information source emits a sequence of symbols from an alphabet that consists of M symbols, which each symbol having a unique corresponding amplitude, so there are M amplitudes for an M-ary system.

When we transmit a symbol, it will have a finite duration of T seconds. The signaling rate of this system is then $1/T$, expressed in symbols per second, or baud. For a binary PAM system, where the value of M is 2, let us define the symbol duration the binary system as T_b seconds, making the system transmit data at a $1/T_b$ bits per second rate. In a $M = 4$ PAM-4 system, the four symbols can be 00, 01, 10, and 11. Therefore, we can see that each symbol represents 2 bits of information, and 1 baud is equal to 2 bits per second in this case. This result can be generalized by saying that one baud is equivalent to $\log_2 M$ bits per second. The symbol duration T of an M-ary PAM system is related to the bit duration T_b of the binary PAM system as:

$$T = T_b \log_2 M$$

In conclusion, an M-ary PAM transmission system can transmit information $\log_2 M$ times faster than the corresponding binary PAM system. If we want to have the same average probability of symbol error, the M-ary PAM system would require more transmitted power. For $M \gg 2$ with an average probability of symbol error small compared to 1, the transmitted power could be increased by a factor of $M^2/\log_2 M$ relative to a binary PAM system.

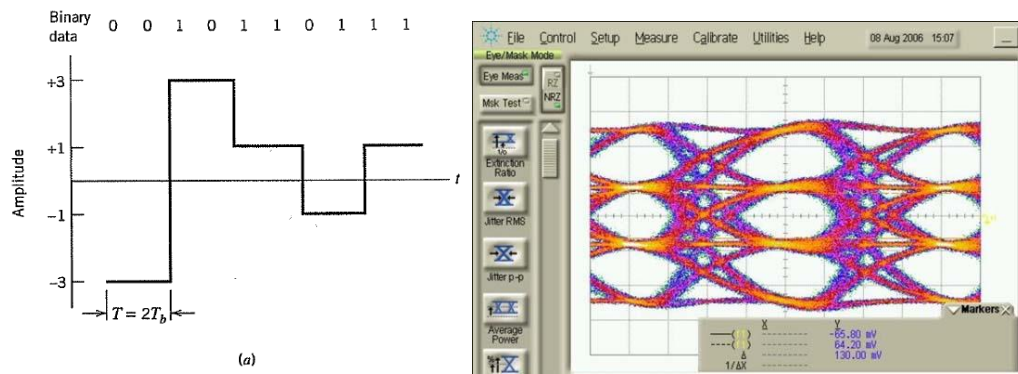


Figure 7.11. Output of a PAM-4 system. (a) Waveform. (b) eye diagrams of practical PAM-4 systems.

7.8 Modulation formats and band-pass transmission of digital signals

PCM system outputs a string of zeroes and ones. These can be directly transmitted as two voltage levels $\pm V$ for a coaxial cable, but need to be modulated in order to be transmitted through space using an antenna.

Transmitting binary data over band-pass communication channels like radio links or satellite channels require the signal to be modulated into a carrier wave that have fixed bandwidth limits set forth by the channel. Stated in earlier chapters, modulation is the process of varying one or more properties of the signal, and in this case, corresponds to switching or keying the amplitude, frequency, or phase of the binary symbols 0 and 1. ASK (amplitude-shift keying), FSK (frequency-shift keying), and PSK (phase-shift keying) are three basic modulation forms.

Figure 7.12(a), (b), and (c), visualize ASK, FSK, and PSK, each respectively.

ASK: Figure 7.12 (a) shows the transmission of 1's as a fixed frequency and amplitude wave for T_b seconds, where as the transmission of 0's are represented by switching off the carrier for T_b seconds. This signal can be generated by taking the on-off form of the binary data and apply it to the carrier using a product modulator.

FSK: Figure 7.12 (b) shows an alternate transmission of the binary data. 1's have one particular frequency, and 0's have a different frequency. This signal can be generated by sending the bipolar form of the input binary data into a voltage controlled oscillator, or switching between two oscillators. The difference between the VCO and oscillator

switching is that VCO shifts the frequency with continuous phase, and oscillator shifts the frequency with discontinuous phase.

PSK: Figure 7.12 (c) shows the transmission of 0's as 180 degrees phase shifted relative to the 1's (sort of like amplitude flipping). This signal can be generated by sending the bipolar form of the input binary data with the carrier to a product modulator.

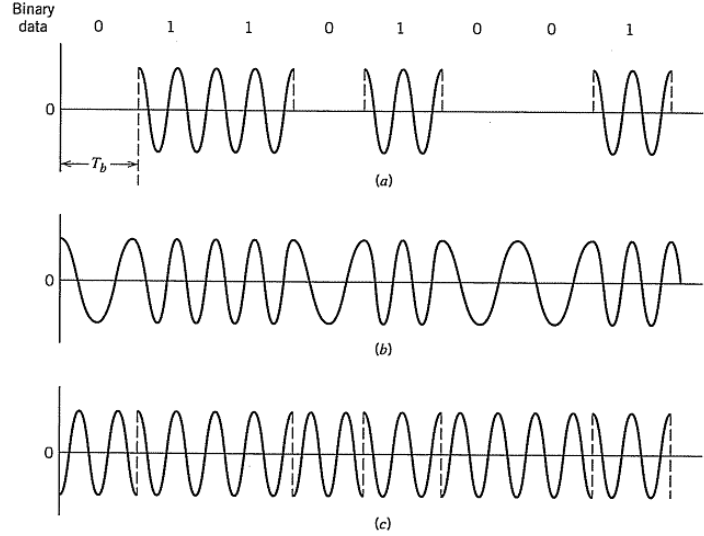


Figure 7.12. Illustration of the three basic forms of signaling binary information: (a) amplitude-shift keying; (b) frequency-shift keying with continuous phase; and (c) phase-shift keying.

Therefore, ASK, FSK, and PSK signals are special (in the sense that they are discrete rather than continuous) cases of amplitude, frequency, and phase-modulated waves. Although it is not easy to see the difference between frequency and phase modulated waves, FSK and PSK is easy to differentiate by comparing the waveforms (see Figure 7.12 (a), and (c)).

7.8.1. Amplitude-shift keying (ASK)

The two modulated signals for binary ASK signals can be described by:

$$\begin{aligned} s_1(t) &= A_c \cos(2\pi f_c t) && \text{for symbol 1} \\ s_0(t) &= 0 && \text{for symbol 0} \end{aligned}$$

7.8.2. Phase-shift keying (PSK)

For BPSK (binary PSK) modulation, the modulation process is shown graphically in Figure 6.14. The binary data is used to create a bipolar line code (Figure 7.13(b)), a representation of the baseband signal $g(t)$. The BPSK signal is then described by:

$$\begin{aligned} s_1(t) &= A_c \cos(2\pi f_c t) && \text{for symbol 1} \\ s_0(t) &= -A_c \cos(2\pi f_c t) = A_c \cos(2\pi f_c t + \pi) && \text{for symbol 0} \end{aligned}$$

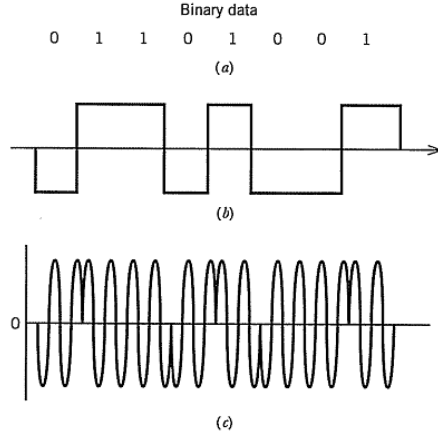


Figure 7.13. The steps of binary PSK modulation: (a) binary data, (b) bipolar line code; and (c) binary PSK waveform.

7.8.3. Frequency-shift keying (FSK)

In FSK, the carrier wave switches its frequency between discrete values in response to the PCM code. BFSK switches its frequency between two discrete values to transmit binary information. In BFSK modulation, complex baseband equivalent signals can't be represented as a line code. FSK transmitters use FM modulators instead of the linear modulation, very much like FM transmitters in Chapter 4 but this time the signal is digital.

The following expressions directly describe a band-pass BFSK signal:

$$s_1(t) = A_c \cos(2\pi f_1 t) \quad \text{for symbol 1}$$

$$s_0(t) = A_c \cos(2\pi f_0 t) \quad \text{for symbol 0}$$

7.9 Complex representation of band-pass signals, M-ary digital signals and QAM

Binary data transmission systems (M -ary system with $M = 2$) can send one signal in the bit interval T_b , either $s_0(t)$ or $s_1(t)$. Similarly, an M -ary data transmission system can send one out of M signals ($s_0(t)$, $s_1(t)$, ..., $s_{M-1}(t)$) in each signaling interval T . Most applications have $M = 2^n$, where n is a natural number, and $T = nT_b$. $s_0(t)$, $s_1(t)$, ..., $s_M(t)$ are called symbols of the system, and the transmission rate of these symbols through a channel is expressed in units of bauds.

We have established that BPSK, and BASK line codes can be expressed in the complex baseband. We can also plot the complex baseband representations geometrically, as shown in Figure 7.14(a) for BASK, and Figure 7.14 (b) for BPSK. M -ary ASK corresponds to M -ary PAM at the complex baseband and is shown in Figure 7.14 (c).

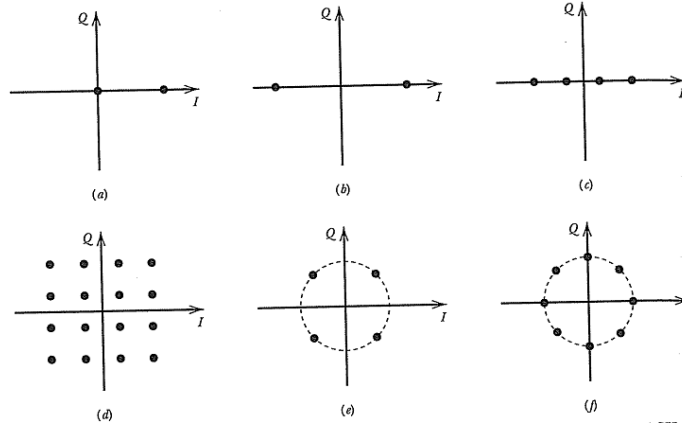


Figure 7.14. Representation of different band-pass modulations in signal space: (a) binary ASK; (b) binary PSK; (c) M -ASK ($M = 4$); (d) M -QAM ($M = 16$); (e) 4-QAM and QPSK; and (f) M -PSK ($M = 8$).

Now consider a modulation scheme that uses independent M -ary PAM sequences for the in-phase and quadrature components, $g_I(t)$ and $g_Q(t)$. We can write the transmitted signal as $\text{Re}\{A_c g(t) e^{j2\pi f_c t}\}$, where $g(t) = g_I(t) + jg_Q(t)$, the *complex baseband equivalent signals*. The complex baseband equivalent is short, compact, and different from the baseband line codes.

This could be represented by the two-dimensional signal space diagram of Figure 7.14(d). This type of band-pass modulation is referred to as quadrature amplitude modulation (QAM). Figure 7.14 (d) illustrates the example of 16-ary QAM (16-QAM). Thus, for this modulation scheme there are $M = 16$ different signals $s_i(t)$, one for each point in the figure, and each signal represents four bits.

If we consider the case of 4-QAM illustrated in Figure 7.14 (e), we observe that all of the points have the same amplitude relative to the origin. A constant amplitude implies that only the phase of the carrier changes so this diagram could also be considered quadrature phase shift keying or QPSK.

This geometric representation of a digital signal as a point in space is referred to as the signal space representation. It is analogous to using a vector, i.e., a phasor, to represent a sinusoid in traditional circuit analysis. The set of points that characterize a given modulation technique is known as the constellation for that modulation.

7.9.1. Quadrature phase-shift keying (QPSK)

In quadrature phase-shift keying, one of four bit pair signals is transmitted every signaling interval. The following example shows how we could represent the four bit pairs:

$$s_0(t) = \sqrt{2}A_c \cos(2\pi f_c t + \pi/4), \quad \text{for bit pair 11}$$

$$s_1(t) = \sqrt{2}A_c \cos(2\pi f_c t + 3\pi/4), \quad \text{for bit pair 01}$$

$$s_2(t) = \sqrt{2}A_c \cos(2\pi f_c t + 5\pi/4), \quad \text{for bit pair 00}$$

$$s_3(t) = \sqrt{2}A_c \cos(2\pi f_c t + 7\pi/4), \quad \text{for bit pair 10}$$

where $0 \leq t \leq T$. We can see that the carrier is transmitted with a phase value (either $\pm\pi/4$ or $\pm3\pi/4$), with each phase value representing a bit pair (see Figure 7.15)

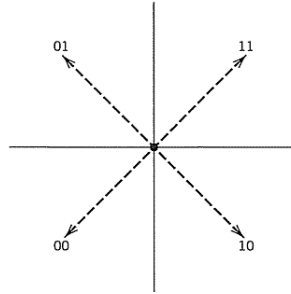


Figure 7.15. illustrating four possible phase values of QPSK signals, with each one corresponding to a unique pair of bits.

Figure 7.16(a) and (b) show a QAM transmitter and receiver block diagram, each respectively. This is identical to the QAM transmitter/receiver in the AM chapter. Just that this time the signals/messages are digital waveforms. The QAM transmitter is made up of two product modulators, an oscillator, and a phase shifter, a signal adder to generate a QPSK signal (2 carrier waves in phase quadrature), as well as a serial to parallel converter which is used to convert the input binary data stream into a parallel format. The serial to parallel conversion makes the signaling interval T of the QPSK signal twice the length of bit duration T_b , which is another way of saying that the transmission bandwidth of the QPSK system would be halved with respect to the corresponding BPSK system, or another way of saying that QPSK can carry twice as many bits with respect to BPSK given the same transmission bandwidth. The QPSK receiver consists of two parallel connected correlators (or detectors). One correlator processes the carrier phase's cosine component, while the other processes the carrier phase's sine component. By processing the signs (positive or negative) of the two correlator outputs, we can see which phase angle was transmitted. A QPSK system can be seen as two parallel BPSK signals with the carrier waves in phase quadrature.

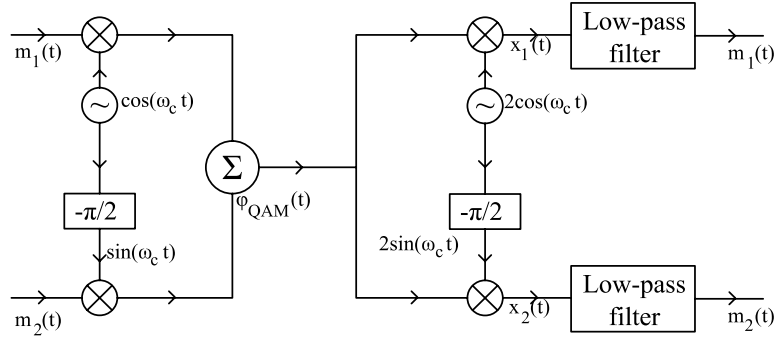


Figure 7.16. QAM transmitter and receiver.

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Real-Time Nyquist Pulse Modulation Transmitter Generating Rectangular Shaped Spectra of 112 Gbit/s 16QAM Signals

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Abstract: A real-time software-defined transmitter generating Nyquist pulses with nearly rectangular spectra is demonstrated at 56 Gbit/s for PDM-QPSK and 112 Gbit/s for PDM-16QAM.

OCIS codes: (060.4510) Optical communications; (060.1660) Coherent communications.

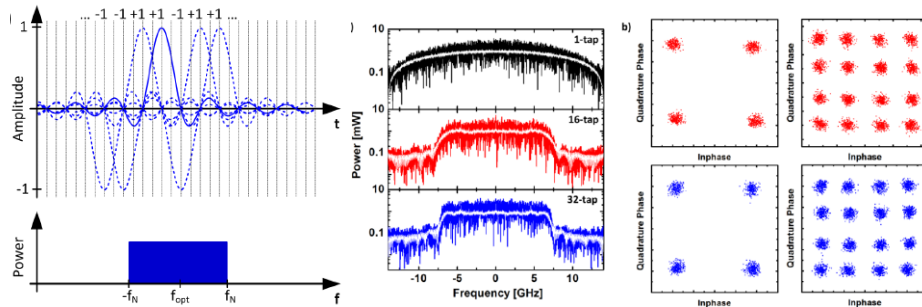


Figure 7.17. 1st Nyquist pulse generation at 14 GBaud with QPSK and 16-QAM modulation formats in 2011 for optical communications. Total bit rate 112 Gb/s.

Optical Arbitrary Waveform Generation and Measurement (OAWG/OAWM) Enabling 320 GBd 32QAM Transmission

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Abstract: We demonstrate optical arbitrary waveform generation at a record-high bandwidth of 325 GHz by phase-stabilized superimposing of four tributary signals. In a proof-of-concept experiment, we use the scheme to generate 320 GBd 32QAM data signals, which are transmitted over 87 km and received by an optical arbitrary waveform measurement system. © 2023 The Author(s)

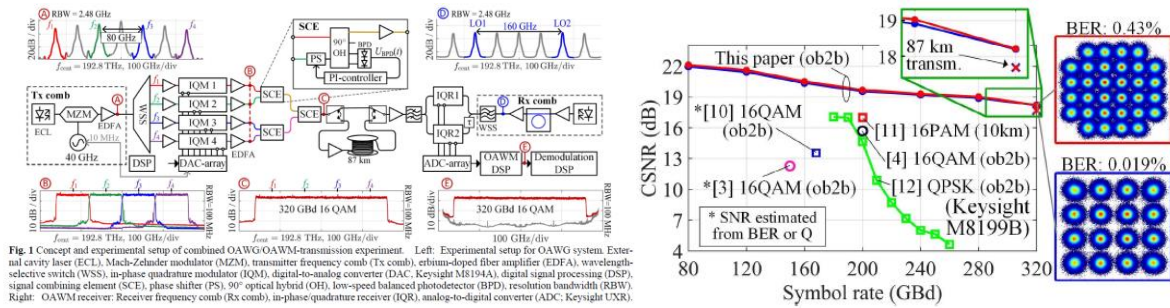


Figure 7.18. The world's highest speed 32-QAM signal generation at 320 Symbol/s (320 GBaud) at 2023 by Karlsruhe Institute of Technology, Germany. The total bit rate is $320 \times \log_2(32) = 1600$ Gb/s. This is generated by 4 separate transmitters and combine the 4 transmitted signals into an overall large bandwidth signal.

M1H.1

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Full C-band 3317-km 12-Coupled-Core-Fiber Transmission with 266.2-Tbps Capacity Using Field-Installed Fiber-Optic Cable

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Abstract: We demonstrate 266.2-Tbps 3317-km full-C-band transmission using field-installed standard-cladding 12-coupled-core fiber cable. MDL impact is effectively suppressed to 0.2 dB per span by using per-core gain control of hybrid MC-EDFA and spatially diverse FEC coding. © 2024 The Authors

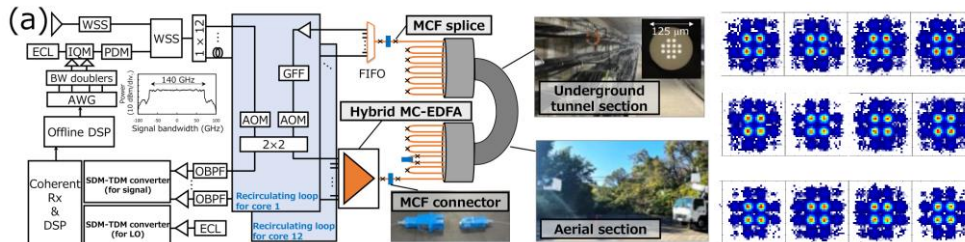


Figure 7.19. The world's fastest speed of 266.2 Tbps across C-band by using 31 WDM 140 GBaud signals in 3317-km 12-core fiber, setting a record decoded capacity-distance product of 889 Tbps×km using only C-band wavelength resources. (a) Experimental setup for field-installed 12-core fiber cable testbed; (b) Constellations at 1544.53 nm for all 12 cores.

440-GBaud All-Electronic Signaling Enabling Single-Wavelength Net Rate over 1 Tb/s per Modulation Dimension

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Abstract: We demonstrate an all-electronic generation of 440-GBaud signals by a 221-GHz digital-band-interleaved DAC. The signals drive a thin-film LiNbO₃ modulator that achieves 440-GBaud single-wavelength PS 16-ASK signaling without optical equalization. © 2025 The Author(s)

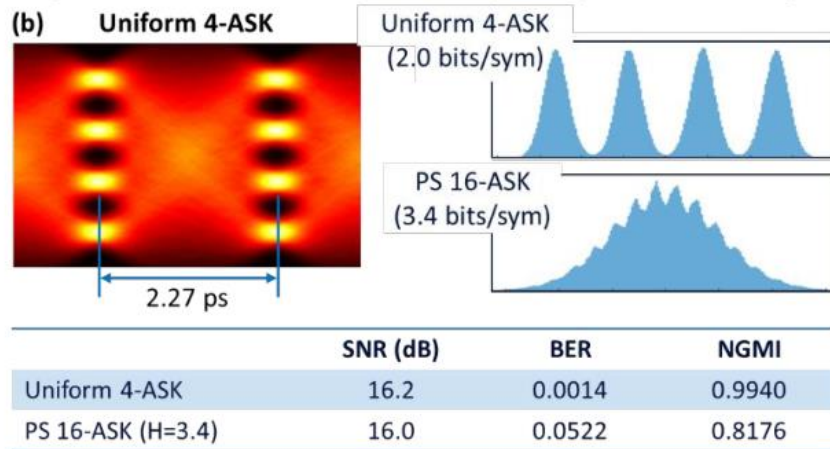


Figure 7.20. 440 GBaud PS-PAM16 (total 1.0447Tb/s), coherent detection, a world record symbol rate in 2025 by using Thin-film LiNbO₃ (Lithium Niobate) modulators, achieved by Nokia Bell Lab in 2025. It is currently holding the world record in highest baud rate transmissions of any modulation format.

Tu.03.06.1

651-Gb/s Net Bitrate IMDD Transmission Using Electrical Bandwidth Multiplexing and Demultiplexing Techniques Based on Ultra-broadband InP-DHBT Mixers

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Abstract We demonstrated 232–264-GBd IMDD signal generation and detection using electrical bandwidth multiplexing and demultiplexing configuration based on in-house 150-GHz InP-DHBT mixers with an adaptive reconstruction technique, achieving a record net bitrate per wavelength of 660-Gb/s back-to-back and 651-Gb/s 11-km transmission with 248-GBd PS-PAM12 signals. ©2025 The Author(s)

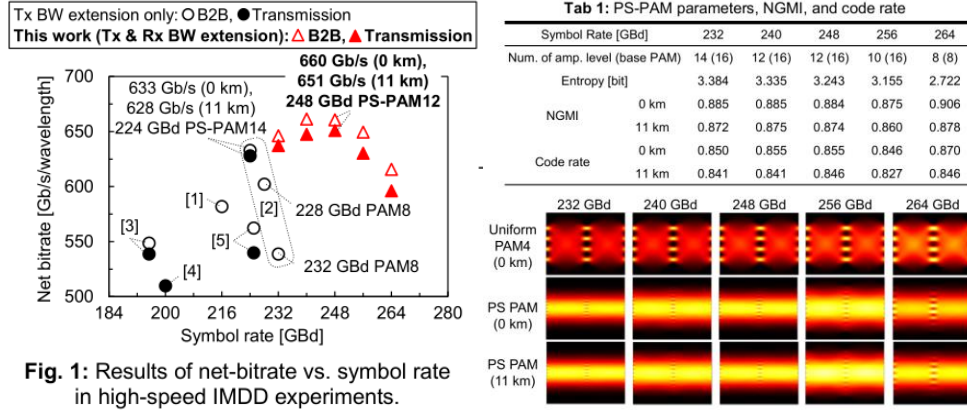


Figure 7.21. For short distance optical links inside data centers in Google and Facebook, we are further limited by cost and component size constraints. Above is transmission over 11 km dispersion-shifted fibre (DSF) with 248 GBd PS-PAM14, reaching a record net bitrate per lambda of 651-Gb/s, published by NTT Japan at ECOC 2025.

A Compact 112-Gbaud PAM-4 Silicon Photonics Transceiver for Short-Reach Interconnects

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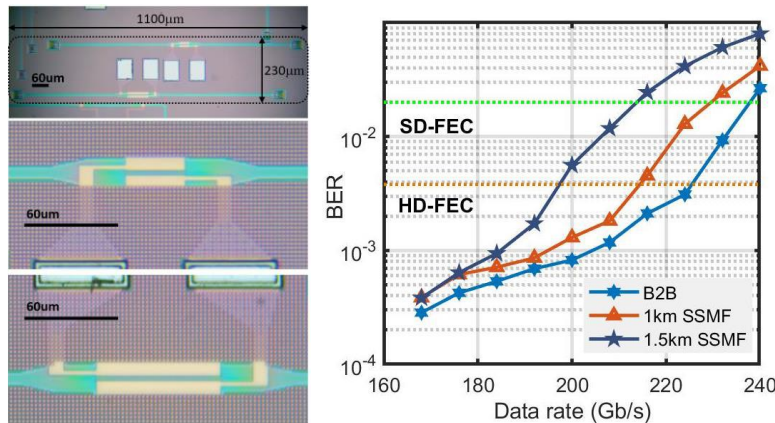


Figure 7.22. For short distance optical links inside data centers in Google and Facebook, we are further limited by cost and component size constraints. Above is 112GSymbol/s PAM-4 (total 224 Gb/s), a world record symbol rate in 2022 by using Germanium Silicon modulators which are ideal for data centers. This is a joint work between PolyU and CUHK.

Power-Fading-Free and IF-Free High-Fidelity Transmission Incorporating Sub-6GHz and mmW Band with Alamouti Coding

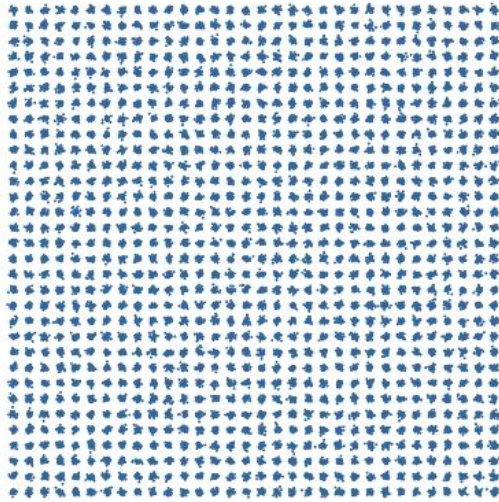
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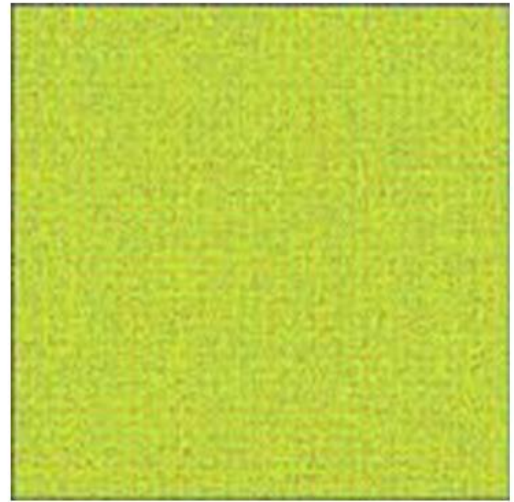
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Abstract We propose a fading-free delta-sigma modulation scheme based on Alamouti coding to overcome dispersion-induced power fading obstacle, and generate dual-band signals at both sub-6GHz and millimeter-wave without up-conversion. We demonstrate high-fidelity transmission of 1048576-/262144-QAM signals at 25-/30-GHz carrier frequencies over 25-/10-km SSMF with direct detection.

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(i) OSR = 6, 1024-QAM
SNR = 38.3dB



(v) OSR = 15, 1048576-QAM
SNR = 67.1dB

Figure 7.23. A fading-free delta-sigma modulation scheme based on Alamouti coding is proposed to overcome dispersion-induced power fading obstacle and generate dual-band signals at both sub-6GHz and millimeter-wave. High-fidelity radio over fiber (RoF) transmission of 1048576-/262144-QAM signals at 25-/30-GHz carrier frequencies over 25-/10-km SSMF with direct detection was demonstrated. The bitrate is 83.3-Gb/s and 90-Gb/s, respectively.

Chapter 8 Noise in Communication Systems

Objectives

- To understand the origins of noise in communication systems
- To characterize noise as a random process
- To familiarize with basic understanding and operations on random variables
- To describe Thermal noise and quantization noise and how it is modeled in communication systems
- To introduce probability of detection error or bit error ratio (BER) and signal-to-noise ratio (SNR)
- To characterize the dependence of BER and system performance on SNR

References: Chapter 5 of S. Haykin and M. Moher, *Communication Systems*, John Wiley and Sons, 5th edition.

The term *noise* is used customarily to designate unwanted waves that tend to disturb the transmission and processing of signals in communication systems and over which we have incomplete control. In practice, we find that there are many potential sources of noise in a communication system. The sources of noise may be external to the system (e.g., atmospheric noise, galactic noise, man-made noise), or internal to the system. The second category includes an important type of noise that arises from spontaneous fluctuations of current or voltage in electrical circuits. This type of noise represents a basic limitation on the transmission or detection of signals in communication systems involving the use of electronic devices. The most common examples of spontaneous fluctuations in electrical circuits is *thermal noise*.

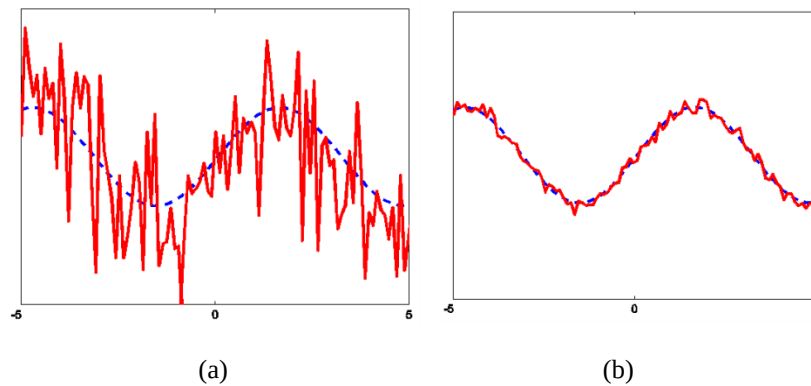


Figure 8.1. (a) With large amount of noise, the received signal will be strongly disturbed. (b) When the noise is small, the received signal will not be as affected, but they are still not identical to the original signal. In communication research, we seek to minimize the amount of noise and/or mitigate the effect of noise through intelligent system design and/or better components/devices.

8.1 Probability, random variables and modeling of noise and information symbols

Since noise is unpredictable, we will model noise as a random quantity. How do we model random things? Typically, we will model noise and other random quantities in the framework of probability and random variables.

8.1.1. Random variables

If we roll a dice, the outcome will be either 1,2,3,4,5 or 6. In a lot of other cases, the outcome may not be a numerical quantity. For example, if we flip a coin, the outcome maybe “head” or “tail”. In these cases, it is often more convenient if we assign a number or a range of values to the outcomes of a random experiment. For the flipping coin example, a “head” could correspond to 1 and a “tail” to 0. In this case, we denote a *random variable* X to describe this process of assigning a number to the outcome of a random experiment. In particular, we will say

$$X = \begin{cases} 0 & \text{with probability } 0.5 \\ 1 & \text{with probability } 0.5 \end{cases}$$

8.1.2. Probability distribution, means and variances of random variables

If a random variable can only take on a discrete set of values (such as rolling a dice or flipping a coin), we call that *discrete random variable*. For thermal noise (which we will talk about in a second) such that it can take on a

continuous range of value, we call that *continuous random variable*. To be a bit more formal, we will define a function $p_x(x)$ as a probability distribution function of the discrete random variable x . For example, in the case of rolling a dice, we can say $p_x(1) = p_x(2) = \dots = p_x(6) = 1/6$. For continuous random variable, the probability distribution $f_x(x)$ of a continuous random variable x is defined such that $P(a \leq X \leq b)$, the probability of X between a and b , is given by

$$P(a \leq X \leq b) = \int_a^b f_x(x) dx.$$

It should be noted that the sum of all probabilities should be 1 i.e.

$$\sum_k p_x(k) = 1 \quad \text{and} \quad \int_{-\infty}^{\infty} f_x(x) dx = 1.$$

For a random variable x , the two quantities that we care about the most is its mean and variance. The mean of x , or the expected value of X $E[X]$, is defined as

$$E[X] = \sum_k k p_x(k) \quad \text{and} \quad E[X] = \int_{-\infty}^{\infty} x f_x(x) dx.$$

Likewise, the variance of x is defined as the mean of the squared distance between the random variable and its mean i.e.

$$\text{Var}(X) = E[X - E[X]]^2 = \sum_k (k - E[X])^2 p_x(k) \quad \text{and} \quad \text{Var}(X) = E[X - E[X]]^2 = \int_{-\infty}^{\infty} (x - E[X])^2 f_x(x) dx.$$

8.1.3. Common random variables

Uniform random variable

A random variable X is said to be *uniformly distributed* over the interval (a, b) if its probability distribution is given by

$$f_x(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

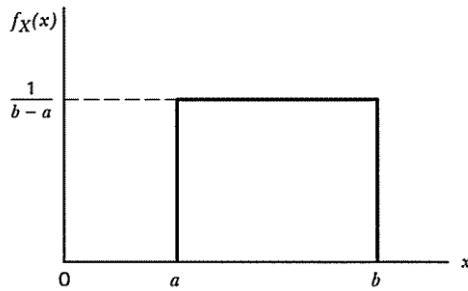


Figure 8.2. Probability distribution for a uniform random variable.

One can show that the mean and variance of this uniformly distributed random variable is given by $(a+b)/2$ and $(b-a)^2/12$ respectively.

Gaussian random variable

A random variable X is said to be *Gaussian distributed* if its probability distribution is given by

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

We sometimes will also write $X \sim N(\mu, \sigma)$ to denote X to be a Gaussian random variable.

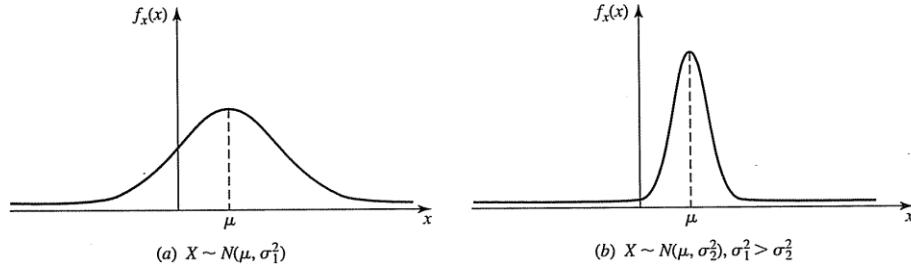


Figure 8.3. Probability distribution for a Gaussian random variable with different variances.

One can show that the mean and variance of this uniformly distributed random variable is given by μ and σ^2 respectively.

8.2 Thermal Noise

Thermal noise is the electronic noise created by the thermal agitation of charge carriers inside a conductor at equilibrium. The mean square value of the thermal noise voltage V_{TN} across a resistor is measured in a bandwidth of Δf Hertz, and is given by:

$$\mathbf{E}[V_{TN}^2] = 4kTR \Delta f \text{ volts}^2$$

while the mean square value of the noise current generator is given by:

$$\mathbf{E}[I_{TN}^2] = \frac{1}{R^2} \mathbf{E}[V_{TN}^2] = 4kT\Delta f/R \text{ amps}^2$$

with k being the Boltzmann constant (approx $1.3806 \times 10^{-23} JK^{-1}$), T is the thermodynamic temperature (or absolute temperature) in Kelvin, and R is the resistance in Ohms. A noisy resistor can be modeled by the Thévenin equivalent circuit which is made up of a noise voltage generator with a mean square value $\mathbf{E}[V_{TN}^2]$ in series with an ideal, noiseless resistor. We could also model a noisy resistor by the Norton equivalent circuit which has a noise current generator parallel to a noiseless conductor. These circuits are shown in Figure 8.4.

Because the resistor electron population is large (large sample size), and the electrons are random in motion and statistically independent of each other, the thermal noise is typically modeled as a normal distribution with a mean of zero.

Because the noise calculations include power transfer, we can also use the maximum power transfer theorem. The maximum power transfer theorem states that the maximum power transferrable from a source to a load is when the internal source resistance is equal to the load resistance. The power from the source is thus divided equally between the internal source resistance, and the load resistance, and the power to the load is coined as "available power". By applying this theorem to the Thévenin and Norton equivalent circuit (see Figure 8.4), the noisy resistor produces an available power of $kT \Delta f$ watts.

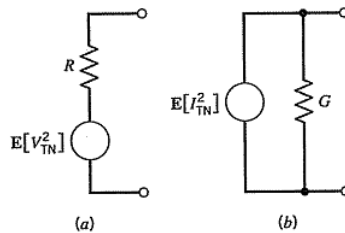


Figure 8.4. Models of a noisy resistor. (a) Thévenin equivalent circuit. (b) Norton equivalent circuit.

8.3 Noise in communication systems and probability of detection error in digital transmission

In a binary digital communication system, the two levels may be represented as an on-off signal, i.e., by an amplitude A and an amplitude 0 as the two possible levels. This is called **unipolar** signaling. Alternatively, one can encode “1” and “0” by equal amplitude pulses of opposite polarities, i.e., by amplitude A and $-A$. The latter is known as a **bipolar** binary signal.

Consider the case bipolar binary signal transmission using an NRZ pulse shape as shown in Figure 8.5. Assume that the binary levels at the receiver are $-A$ and A , i.e., the transmitted signal $g(t)$ is as shown. The thermal noise in all parts of the electronic devices in the transmitter/receiver is collectively modeled as $n(t)$ introduced at the receiver. This is a very common way to model thermal noise in electronic circuits and we call this noise additive white Gaussian noise (AWGN). (We will explain what it means by “white” if time allows) The waveform $s(t)$ at the receiver is given by also contains random noise $n(t)$, so that

$$s(t) = g(t) + n(t)$$

$s(t)$ is then sampled at regular intervals of T (bit period). The samples Y will then be

$$Y = \begin{cases} A + n(kT) & \text{bit '1' is sent} \\ -A + n(kT) & \text{bit '0' is sent} \end{cases}$$

for $k=1,2,3,\dots$. In this case, the detection threshold should obviously be 0 : that is, if $Y > 0$, the bit is detected as “1”; if $Y < 0$, the bit is detected as “0”.

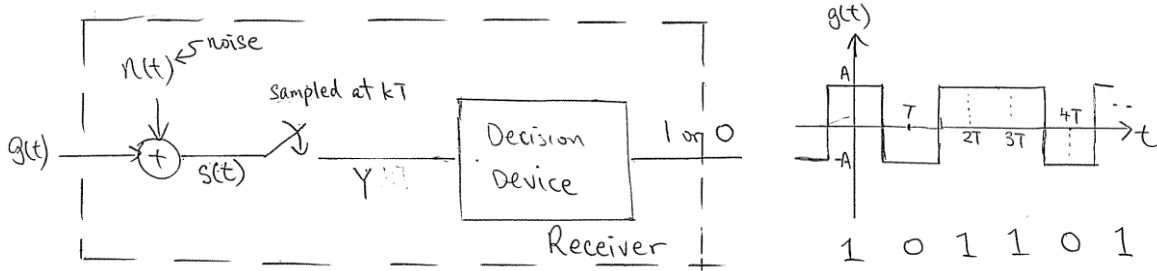


Figure 8.5. Modeling of noise in a digital communication system. The thermal noise in all parts of the electronic devices in the transmitter/receiver is collectively modeled as $n(t)$ introduced at the receiver.

Now, let the noise component in Y be modeled as a Gaussian random variable with mean of 0 and variance σ^2 such that

$$f_{n(kT)}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}.$$

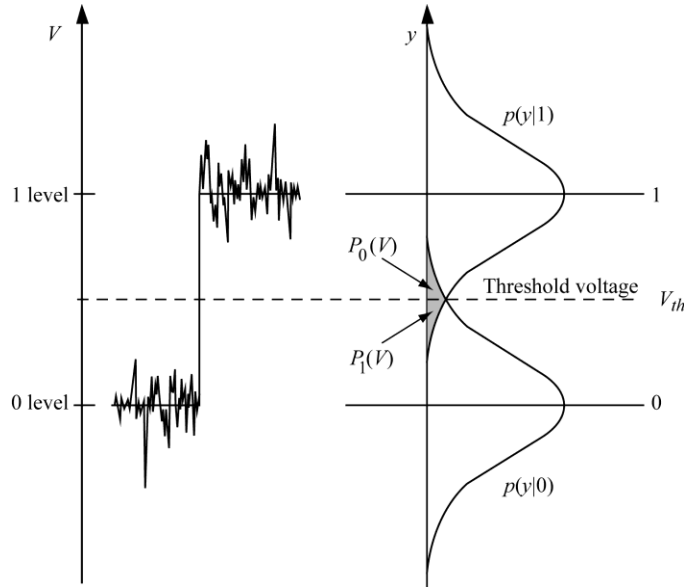


Figure 8.6. Probability distributions for 0 and 1 signal levels.

In this case, referring to Figure 8.6, there is a probability that the noise amplitude can exceed that of the signal i.e. when the $A + n(kT)$ is negative or $-A + n(kT)$ becomes positive. In this case, a bit '1' will be detected as bit '0' and a bit '0' will be detected as bit '1' respectively. This is called a **bit detection error**.

8.3.1. Probability of detection error in baseband binary communication systems

For bipolar signaling with amplitude A and $-A$ and thermal noise with mean 0 and variance σ^2 , detection errors occurred when "1" is interpreted at the receiver as a "0" and vice versa. We denote the probability of these events to be $P(0|1)$ and $P(1|0)$ respectively. In particular, the probability distribution of Y will be given by

$$f_Y(y) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-A)^2/2\sigma^2} & \text{bit '1' is sent} \\ \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y+A)^2/2\sigma^2} & \text{bit '0' is sent} \end{cases}$$

In this case, detection error will occur with probability

$$P(0|1) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^0 e^{-(y-A)^2/2\sigma^2} dy \quad P(1|0) = \frac{1}{\sqrt{2\pi}\sigma} \int_0^{\infty} e^{-(y+A)^2/2\sigma^2} dy$$

The overall probability of detection error will then be given by

$$P_e = P(1) P(0|1) + P(0) P(1|0)$$

Where $P(0)$ and $P(1)$ are the probability of '0' and '1' being sent respectively i.e. the relative frequencies of occurrence of 0 and 1 respectively. Typically, we can safely assume $P(0) = P(1) = 1/2$. Also, from simple symmetry arguments, one can deduce that $P(0|1) = P(1|0)$. Therefore,

$$P_e = 0.5 P(0|1) + 0.5 P(1|0) = P(0|1).$$